How much of the memory of face-to-face contact dynamics is network context?

Jobst Heitzig*
Potsdam Institute for Climate Impact Research, Member of the Leibniz Association, 14473 Potsdam, Germany
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Durations of face-to-face contacts and of the intervals between them are broadly distributed, which is usually explained by an intrinsic memory: the longer a contact lasts, the less likely it is to end. An alternative explanation is that the dynamics is Markovian in a richer state, in which the surrounding network (common contacts, the numbers of current contacts, friendships, classes) and decaying traces of recent interactions modulate the rates. We compare both explanations by exact likelihoods on the contacts of 329 students of a high school over five days (53 224 contacts), using a simulator and inference engine for typed multi-relational network dynamics in which all candidate models are specified as rule sets. Models are fitted on three days and evaluated on two held-out days by their log-likelihood and by time-rescaling residuals. Memory dominates: decaying traces of recent contacts raise the held-out log-likelihood by 0.91 per event, while network context adds 0.22 per event and absorbs only 8% of the memory effect. A Markovian model with traces on four time scales and network context outperforms a non-Markovian model with power-law aging on the held-out days, but only with network feedback that is dynamically unstable; among stable models, power-law aging predicts best. Latent heterogeneity between pairs explains a part, but not all, of the advantage of power-law aging. All fitted models with network feedback are dynamically unstable: simulated forward, the network becomes dense within minutes, whereas the data stay sparse. A pair closure generated from the same rules predicts this instability, which a standard mean-field closure misses by an order of magnitude in time. Constraining the feedback to the stable region costs 0.12 per held-out event and retains 45% of the improvement by network context.

I. INTRODUCTION

Wearable proximity sensors record face-to-face contacts between people at a resolution of seconds [1, 2], and the resulting temporal networks [3] have become a standard empirical basis for models of social contact and epidemic spreading. Their most robust statistical feature is the broad distribution of contact durations and of the intervals between successive contacts of the same pair, far from the exponential distributions of a memoryless process. The prevailing explanation is an intrinsic memory of the individuals: the probability to leave a contact or a group decreases with the time already spent in it, as in the reinforcement models of Zhao *et al.* [4] and Starnini *et al.* [5], or ties are reinforced by repeated interaction, as in activity-driven networks with memory [6]. Such aging makes the process non-Markovian in the network state.

An alternative explanation is that the process is Markovian, but in a richer state than the set of current contacts. A contact embedded in a group (with common contacts), between members of the same class, between friends, or of a student who currently has many other contacts, may have very different rates of ending than an isolated contact between strangers, and a mixture of such rates, changing as groups form and dissolve, also produces broad durations. Traces of recent interactions that decay over time add memory to such a description while keeping it Markovian in an extended state, since the traces are part of the state. The two explanations have different consequences: intrinsic aging puts the memory into the individuals, network context puts it into the configuration around them, and only the latter is captured by models of network structure.

Here we quantify how much of the memory is network context. We specify a hierarchy of Markovian models with increasing context and memory, and a competing non-Markovian aging model, for the contacts of 329 students of a high school [7], fit all models by maximum likelihood on three days, and compare them on two held-out days by their log-likelihood and by time-rescaling residuals [8]. The Markovian models are rule sets for tricl, a simulator and inference engine for typed multi-relational network dynamics, which computes the exact log-likelihood and its gradient for each rule set, including rates that decay between events.

We find that memory dominates and that network context explains only a small part of it; a Markovian model with decaying traces on several time scales and network context outperforms power-law aging, but only with unstable network feedback. We also find that all fitted models with network feedback are dynamically unstable, a form of model degeneracy that a pair closure of the generated mean-field equations predicts and a standard closure misses. Section II describes the data, Sec. III the models, Sec. IV inference and evaluation, Sec. V the results, Sec. VI the stability analysis and closures, and Sec. VII the conclusions.

II. DATA

The SocioPatterns collaboration recorded face-to-face proximity between the students of nine classes of a high school in Marseille from Monday noon to Friday after-

* heitzig@pik-potsdam.de

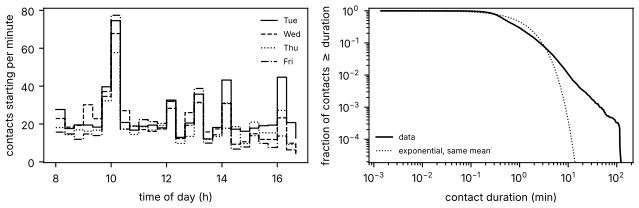


FIG. 1. Left: contacts starting per minute over the school day, for the four full days. The peaks at the start of each hour and at the morning break reflect the timetable. Right: distribution of contact durations, compared with an exponential distribution of the same mean.

noon in December 2013, in 20-second intervals [7]. We use the version distributed by the Netzschleuder repository (`sp_high_school`), with 329 students in 9 classes, 1 437 Facebook friendships among them, and 668 friendship nominations from a survey. Consecutive records of a pair with gaps of at most 40 seconds form one contact, which gives 53 224 contacts; with gaps of up to 120 seconds there are 36 277, so that 32% of the contacts are separated from the next one of the same pair by only one to five missing records, and we report the sensitivity of all results to this choice (Sec. V D). The start and end of each contact are drawn uniformly within the first and last 20-second interval, which removes ties without favoring any order. Of all contacts, 93% are between students of the same class; the median duration is 0.5 minutes, and 1.1% of the contacts last ten minutes or longer (Fig. 1).

The timetable is visible in the data as peaks of contact formation at the start of each hour and during the morning break (Fig. 1, left). It is an exogenous driver, which we include in all models as forced events: at the start of each hour, all students are linked to a hub entity for that hour, during the first 20 minutes of each hour to a hub entity for the change of rooms, and from 10:00 to 10:20 to a hub entity for the morning break. Rates depend on these links through the numbers of links of each student to the hub entities, and the likelihood is conditional on the forced events. Pairs that met on an earlier day are linked by a relationship that is also set by forced events at the start of each day. We use Monday to Wednesday (65 502 events) for estimation and Thursday and Friday (40 946 events) for evaluation.

III. MODELS

A. Model class

All Markovian models are rule sets of one class, in which a state is a set of typed links between typed entities and every link is established and terminated at rates that depend on the configuration around it. For a candidate link ℓ between students i and j, the rate is

$$\lambda(\ell, t) = A \operatorname{expit}\big(4B(\ell, t)\big), \qquad B = b_0 + \sum_k b_k \, n_k(\ell, t), \tag{1}$$

where A is a ceiling rate, $\operatorname{expit}(x) = 1/(1+e^{-x})$, and the n_k count configurations of type k adjacent to ℓ: *angles*, i.e. two-paths $i \to m \to j$ of given relationship types (common current contacts, being in the same class, or, through the identity relationship, a friendship of the pair itself), and *legs*, i.e. links of one endpoint of given types (the current contacts of i or of j, or the link of i to the hub entity of the current hour). As long as $\operatorname{expit}(4B)$ is small, the rate is log-linear in the counts, $\lambda \approx Ae^{4B}$, as in relational event models. Contacts are links of a symmetric relationship type, so that the two directions of a pair are two candidate events with equal rates, and a pair forms or ends a contact at twice the rate of Eq. (1). We use $A = 1$ per minute for formation and $A = 6$ per minute for termination, ceilings that are reached only for pairs with very strong positive effects.

Memory enters through *actions*: links whose *impact* jumps by one at every act and decays exponentially at a rate δ, and which cease to exist when the impact falls below a threshold (here 0.01, a part of the model definition). An act of the action type $\texttt{started}_\delta$ happens whenever a contact starts, and one of $\texttt{ended}_\delta$ whenever a contact ends, so that the impact of the pair's own action, $e^{-\delta\tau}$ summed over its recent starts or ends, encodes the time τ since these events. An angle through an action counts with its impact instead of one, so that b_k times the impact of $\texttt{started}_\delta$ in the termination rate makes the ending hazard depend on the age of the contact, and the corresponding term with $\texttt{ended}_\delta$ in the formation rate makes the re-formation rate depend on the time since the last contact of the pair. With several decay rates, these terms approximate an arbitrary dependence on the age by a sum of exponentials, while the process remains Markovian in the extended state of links and impacts.

B. Model hierarchy

All models include the timetable (a base value and effects of the hour, the change of rooms and the morning break, separately for formation and termination).

M0 Timetable only.

M1 M0 plus static context: same class, Facebook friendship, survey friendship in either direction.

M2 M1 plus network context: the number of common current contacts (triadic closure and group stabilization) and the numbers of current contacts of both students (not counting the pair's own contact).

M4, M6 M1 plus memory: met on an earlier day, and the impacts of $\texttt{started}_\delta$ (in termination) and $\texttt{ended}_\delta$

(in formation) for $\delta \in \{0.5, 0.05\}$ per minute (M4) or $\delta \in \{2, 0.5, 0.1, 0.02\}$ per minute (M6).

M3, M5 M2 plus the memory terms of M4 and M6, respectively.

Every effect enters both directions of a pair with the same coefficient, so that the rates are invariant under exchanging i and j; then the log-likelihood of the event sequence with recorded directions differs from the one without by a constant (App. A).

C. Competing model: intrinsic aging

The competing model C1 has the timetable and static context of M1, met on an earlier day, and an intrinsic aging of the form of Zhao *et al.* [4] and Starnini *et al.* [5] at the level of pairs: the probability units of termination contain $\beta_T \ln(1+\tau/\tau_0)$ with τ the age of the contact, and those of formation contain $\beta_F \ln(1+\sigma/\tau_0)$ with σ the time since the pair's last contact ended (or a separate constant if the pair has not met that day), so that far from saturation the rates are power laws $(1+\tau/\tau_0)^{4\beta_T}$ and $(1+\sigma/\tau_0)^{4\beta_F}$. These rates depend on ages that are not part of the network state, so C1 is not Markovian in it. C1 has no network context; its likelihood is computed by a separate implementation with the same conventions, which agrees with tricl to 10^{-12} when the aging terms are switched off. The scale τ_0 is chosen among $\{1/3, 1, 3\}$ minutes by the training likelihood (best: 3 minutes).

D. Latent heterogeneity

In survival analysis, unobserved heterogeneity mimics aging: if some pairs have long contacts and others short ones, the hazard of the pooled contacts falls with duration although no pair's hazard does. We test this explanation with latent affinity classes of pairs, which can be thought of as hidden static relationship types: each pair belongs to one of K classes, which add class-specific constants to the probability units of formation and termination and have prior probabilities π_k. For the models whose rates depend only on the pair's own history, the static context and the timetable (M6 and C1), the likelihood factorizes over pairs, and the marginal likelihood $\prod_{\text{pairs}} \sum_k \pi_k L_k(\text{pair})$ is exact; we maximize it with analytic gradients, and a separate implementation reproduces tricl's log-likelihood of M6 when $K = 1$. On the held-out days, the class probabilities are either the prior ones or each pair's posterior probabilities given its training days.

IV. INFERENCE AND EVALUATION

A. Exact log-likelihood

For a fully observed sequence of events in a window $[0, T]$, the log-likelihood is

$$L(\theta) = \sum_i \ln \lambda_{\tau_i}(\ell_i, t_i^-; \theta) - \int_0^T \Lambda(t; \theta)\, dt, \qquad (2)$$

where Λ is the total rate of all possible events. tricl maintains Λ incrementally during a replay of the observed events, including the pairs without any adjacent angle, whose rates depend only on the degrees of their endpoints and which are summed through histograms of degree vectors. Rates with decaying impacts vary continuously between events; their integrals are accumulated per event by Gauss–Legendre quadrature whenever the event's rate is updated (App. A). Forced events and acts triggered by contact events add no terms, so that L is conditional on the timetable. tricl also computes the gradient of L with respect to all coefficients, and we maximize L with L-BFGS-B [9], summing over the training days, which are independent observation windows (the network is empty at the start of each day).

B. Held-out likelihood and time rescaling

We compare models by their log-likelihood on the held-out days, evaluated at the estimates from the training days, which penalizes overfitting directly. As a goodness-of-fit test, we use the time-rescaling theorem [8]: if the model is correct, the integrated termination hazard Λ of each contact from its start to its end is a unit exponential variable, independently across contacts. tricl outputs these integrals for the observed contacts, summed over both directions of a pair and including time-varying rates; we report the Kolmogorov–Smirnov distance of their distribution from the unit exponential distribution. On simulated data from a known model with impacts and a timetable, the integrals are indeed unit exponential variables (a test of the implementation).

C. Simulation

tricl simulates the models exactly with the direct method over a sum tree of rates; events with decaying rates are simulated by thinning [10, 11], with upper bounds that are valid until the next update of each event, and forced events and the expiry of actions are processed at their times (App. B).

V. RESULTS

A. Held-out likelihood

Table I reports the log-likelihoods on the training and held-out days. Static context improves the held-out log-likelihood by 0.99 per event over the timetable alone, mostly through the strong preference for classmates. Network context adds 0.22 per event (M2 over M1). Memory adds 0.91 per event without network context (M6 over M1) and 0.84 per event with it (M5 over M2), so that network context absorbs only 8% of the memory effect, while it still adds 0.15 per event when memory is present (M5 over M6). With two time scales of memory, the Markovian model M3 falls short of the aging model C1 by 3 583 log-likelihood units on the held-out days; with four time scales, M5 exceeds C1 by 556 units. A Markovian description with network context and traces on several time scales therefore predicts held-out contacts at least as well as intrinsic power-law aging; however, its network feedback is dynamically unstable (Sec. VI), and among the stable models (last column of Table I) C1 predicts best, 5 517 units better than the Markovian memory model M6 without network context.

B. Estimated effects

Table II shows the estimates of the best model M5 (probability units; a coefficient b multiplies the rate by about e^{4b} per unit of its count). Classmates form contacts about 34 times as often as other pairs, and each common current contact multiplies the formation rate by about 21, so that contacts form predominantly within existing groups. Common contacts lower the termination rate (groups stabilize contacts), whereas more current contacts of either student raise both rates. The memory terms combine to a termination hazard that is raised during the first few minutes of a contact and lowered afterwards, and to a re-formation rate that is suppressed during the first half minute after a contact ended (which the definition of contacts by gaps of at least 40 seconds excludes) and strongly raised during the following minutes. In C1, the corresponding power-law exponents are -1.29 for termination and -1.29 for re-formation.

C. Time-rescaling residuals

No model passes the time-rescaling test, but the residuals improve in the same order as the held-out likelihood (Table I, Fig. 2): the KS distance falls from 0.180 for M1 to 0.090 for M5, while C1 reaches 0.119. The rescaled durations are too rare near zero and too frequent between about 0.1 and 0.4: the models overestimate the hazard at the very start of a contact and underestimate it at intermediate ages.

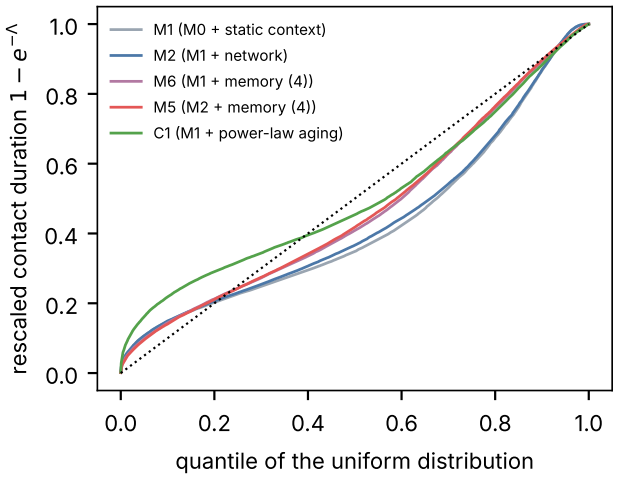


FIG. 2. Time-rescaling diagnostic on the held-out days: quantiles of $1 - e^{-\Lambda}$, with Λ the integrated termination hazard of each contact under the fitted model, against the quantiles of the uniform distribution. A correct model lies on the diagonal.

D. Sensitivity to the definition of contacts

Merging records with gaps of up to 120 seconds instead of 40 seconds into one contact reduces the number of contacts by a third and changes the size, but not the order, of the effects (Table III). Memory still dominates, but its gain without network context falls from 0.81 to 0.62 per event (two time scales, M4 over M1), so that a part of the short-term memory reflects contacts that are interrupted by detection failures rather than ended. Network context gains 0.27 per event with the longer gaps. The advantage of the aging model shrinks more: with the longer gaps, C1 gains 0.72 per event over M1 and falls behind the Markovian model M3 with two memory time scales (0.83 per event), so that a part of the apparent power-law aging also reflects interrupted contacts; it remains better than the stable Markovian model M4 by 0.09 per event.

E. Latent heterogeneity

Latent affinity classes of pairs improve both pair-local models, but the Markovian memory model M6 gains considerably more than the aging model C1: with $K = 4$ classes, the held-out log-likelihood rises by 6 897 units for M6 and 4 247 units for C1 (Table IV). The advantage of C1 over M6 therefore shrinks from 5 517 to 2 868 units, and M6 with latent classes predicts the held-out days better than C1 without them (by 1 380 units). A part of the apparent aging is thus unobserved heterogeneity of pairs, but not all of it, since the remaining gap hardly changes between two and four classes. The classes are not persistent properties of pairs: using each pair's posterior class probabilities from the training days lowers the held-out

TABLE I. Log-likelihoods L on the training days (Monday to Wednesday) and the held-out days (Thursday and Friday), held-out improvement per event relative to M1, the Kolmogorov–Smirnov distance (KS) of the time-rescaled contact durations on the held-out days from the unit exponential distribution (zero for a correct model), and whether simulations of the fitted model stay sparse (Sec. VI).

model	components	param.	L (training)	L (held out)	ΔL/event	KS	stable
M0	timetable	22	−368 997	−235 337	−0.987	0.186	yes
M1	M0 + static context	28	−302 346	−194 943	+0.000	0.180	yes
M2	M1 + network	32	−288 045	−186 055	+0.217	0.163	no
M4	M1 + memory (2)	34	−253 614	−161 772	+0.810	0.112	yes
M3	M2 + memory (2)	38	−243 554	−155 741	+0.957	0.111	no
M6	M1 + memory (4)	38	−247 207	−157 675	+0.910	0.102	yes
M5	M2 + memory (4)	42	−237 010	−151 601	+1.058	0.090	no
C1	M1 + power-law aging	33	−240 850	−152 158	+1.045	0.119	yes

TABLE II. Estimates (probability units) and standard errors (observed information) of the network-context and memory effects of model M5 on the formation and termination rates of contacts. The timetable effects are not shown.

	formation		termination	
effect	estimate	s.e.	estimate	s.e.
same class	0.882	0.006	−0.001	0.006
Facebook friends	0.062	0.004	−0.013	0.004
survey friendship	0.070	0.003	−0.017	0.002
common contact	0.762	0.009	−0.186	0.005
other current contact	0.112	0.002	0.036	0.002
met on earlier day	0.148	0.004	−0.008	0.003
memory, $\delta = 2$/min	−2.790	0.034	−0.504	0.010
memory, $\delta = 0.5$/min	1.529	0.019	0.699	0.010
memory, $\delta = 0.1$/min	0.042	0.007	−0.129	0.004
memory, $\delta = 0.02$/min	0.087	0.001	0.002	0.001

TABLE III. Sensitivity to the definition of contacts: held-out log-likelihood and improvement per event relative to M1 with gaps of up to 120 seconds merged into one contact, and the improvement per event with gaps of up to 40 seconds (Table I) for comparison.

model	L (held out)	ΔL/event	ΔL/event, 40-s gaps
M0	−175 221	−0.955	−0.987
M1	−148 269	+0.000	+0.000
M2	−140 726	+0.267	+0.217
M4	−130 737	+0.621	+0.810
M3	−124 936	+0.827	+0.957
C1	−128 071	+0.715	+1.045

log-likelihood (by 770 units for M6 with $K = 4$) compared with the prior probabilities, so the heterogeneity that matters varies from day to day. The smallest class comprises about 2% of the pairs.

TABLE IV. Latent affinity classes of pairs for the pair-local models M6 and C1: log-likelihoods on the training days and on the held-out days, with the prior class probabilities or with each pair's posterior class probabilities given its training days.

model	K	param.	L (train)	L (held out)	L (posterior)
M6	1	38	−247 207	−157 675	–
M6	2	41	−239 952	−152 295	−153 500
M6	3	44	−238 164	−151 243	−152 142
M6	4	47	−237 268	−150 778	−151 548
C1	1	33	−240 850	−152 158	–
C1	2	36	−236 563	−149 245	−149 541
C1	3	39	−235 366	−148 563	−148 670
C1	4	42	−234 022	−147 911	−147 928

VI. STABILITY AND CLOSURES

A. Instability of fitted network feedback

The fitted models are evaluated above on the observed paths, where the network is sparse: on average only 23 contacts within classes exist at a time. Simulated forward from an empty network with the timetable of a held-out day, every model with network context (M2, M3, M5) behaves differently: within minutes, contacts form faster than they end, and the network becomes dense (Fig. 3). The same happens when common contacts raise the attempt rate additively instead of the probability units (a variant with linear instead of exponential triadic closure, whose held-out likelihood is close to that of M2). The reason is a positive feedback that the likelihood does not constrain: each common contact and each further current contact raises the formation rate, in the observed sparse states moderately, but without bound as groups grow, and the stabilizing effect of common contacts on termination adds to it. Models without network context (M1, M4, M6, C1) are stable. This is a form of model degeneracy, analogous to the degeneracy of exponential

random graph models with triangle terms [12]: the fitted parameters describe the observed configurations, but the process they define concentrates on configurations that are never observed.

B. Simple and pair closure

Mean-field equations generated from the rules can anticipate such behavior without simulation, but only if the closure keeps the correlations that drive it. We compare two closures for the contacts within classes of the fitted M2 (93% of all contacts), treating each class as a homogeneous block whose pairs differ by their static marks (Facebook and survey friendship). The *simple closure* takes the numbers of common contacts and other contacts of a pair as Poisson variables with means given by the current density, independently of whether the pair is in contact. The *pair closure* adds as state variables the sums Z and O of the common contacts over pairs in contact (Z = three times the number of triangles) and over pairs not in contact (open two-paths); given the state of a pair, its common contacts are Poisson with mean Z/L or O/U (L, U the numbers of pairs in and not in contact), and the excess degrees follow from the degree moments implied by $Z + O$, the number of two-paths (App. D). Both closures have the same dense stationary state of about 791 contacts within classes, but they reach it at very different times (Fig. 3): the pair closure tips after about 14 minutes, close to the simulation (about 14 minutes), the simple closure only after about 142 minutes. Triadic feedback acts through the correlation between a pair's state and its common contacts, which the simple closure discards; the pair closure keeps it, and with it the time scale of the instability.

C. Stability-constrained fit

A stable model with network context can be obtained by maximizing the likelihood under upper bounds on the destabilizing coefficients and checking the stability of each constrained fit. Bounding only the effect of common contacts on formation does not suffice: the fit then compensates through the effect of current contacts, and even with the bound 0.30 (instead of the free estimate) simulations reach 67 times the largest observed number of simultaneous contacts, although the pair closure, which ignores fluctuations and contacts between classes, predicts no tipping already for the bound 0.47. With both effects bounded, the largest bound on the effect of current contacts for which single simulations on the training days stay sparse is 0.046, but repeated simulations of the held-out days still tip occasionally (up to 23 times the observed maximum), since stability near the boundary is a matter of fluctuations. With the bound zero (no positive effect of current contacts on formation, instead of the free estimate 0.10) and the bound 0.47 on common contacts

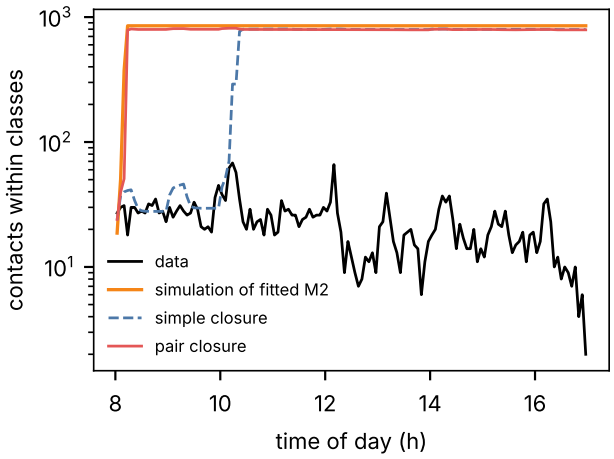


FIG. 3. Contacts within classes over a held-out day: data, simulations of the fitted network-context model M2 (mean of three runs, stopped when the network had become dense), and the simple and pair closures of the same model, all started from an empty network at 8:00 with the timetable as forced events.

(each common contact multiplies the formation rate by at most 6.6), all simulations stay sparse (at most 1.08 times the observed maximum). This stable model costs 4 919 log-likelihood units on the held-out days (0.12 per event) relative to the unconstrained M2 and retains 45% of the improvement of network context over M1. Stability thus constrains how strongly the network context may reinforce itself, and the remaining network effect is a lower bound on what a stable Markovian description can attribute to it. With the same bounds, the model with network context and memory (M5) is still unstable, since recent contacts add a feedback of their own, and it falls short of C1 by 2 568 units on the held-out days.

VII. DISCUSSION

The memory of face-to-face contact dynamics is not primarily a property of the surrounding network. Static context (classes, friendships) is the largest single source of predictability, but among the dynamic explanations, decaying traces of a pair's own recent contacts improve the prediction of held-out days about 0.91 log-likelihood units per event, network context only 0.22, and network context absorbs only 8% of the memory effect. Whether the memory requires intrinsic aging outside the network state depends on stability: traces on four time scales, which keep the process Markovian in an extended state, predict the held-out days better than power-law aging when combined with unconstrained network context, but this combination is dynamically unstable, and among stable models power-law aging predicts best. A part of the short-term memory reflects the definition of contacts from sensor records, since merging longer gaps reduces it

(Sec. V D); this is a caveat for all models of such data, including the aging models whose motivation is the broad distribution of durations.

The instability of the fitted network models is a second result with wider implications. Likelihood fits of rates that grow with common contacts and current contacts are well determined by the sparse configurations that are observed, but the process they define is degenerate. Simulation studies with such fitted models would therefore show runaway group formation, and predictive checks by simulation, not only by likelihood and residuals, are necessary for network-feedback models. The pair closure is a cheap diagnostic of this degeneracy, which the standard closure misses, but it is too optimistic to replace simulations as a stability criterion, since it ignores fluctuations; constrained maximum likelihood with a simulation check yields a stable model that keeps 45% of the network effect. A representation that bounds group sizes, e.g. contacts as joint membership in group entities whose join and leave rates depend on the group size, removes the unbounded feedback by construction; we pursue this in ongoing work, together with a comparison of the predicted distributions of durations and group sizes by simulation.

The comparison was possible because all Markovian models are rule sets of one model class for which tricl computes exact likelihoods, gradients and residuals, including rates that decay between events, exogenous forced events, and acts triggered by contact events. The same machinery applies to other multi-relational temporal networks with fully observed event sequences, such as communication and co-location data, where decaying impacts of calls and messages and their coupling to co-location can be compared in the same way.

CODE AND DATA AVAILABILITY

tricl is free software under the GNU General Public License v3 at `https://github.com/pik-gane/tricl`. The scripts in its directories `study/` and `paper/` download the public SocioPatterns data from the Netzschleuder repository, fit all models, and regenerate every figure, table and number of this paper.

USE OF AI TOOLS

Tools. This work used Anthropic's Claude Code (cloud sessions, September–November 2026) with the models [MODELS], including parallel sub-agents of the same models.

Substantive use. Under the author's direction, the AI system wrote most of the current version of tricl (including leg influences, actions with decaying impacts, forced events, triggered acts, the replay mode with analytic gradients, time-rescaling residuals, and their tests) and the scripts of this study (data processing, model specifications, fitting, the competing aging model, residuals, closures, figures and tables). It ran the fits and simulations reported here, searched and checked the literature, and drafted the text of this paper.

Assistive use. Copy-editing, consistency checks between text, code and generated numbers, and checking of bibliographic data against Crossref.

Direction and verification. The author designed the model class and wrote the original simulator, set the requirements for the study (an application with Markovian dynamics, a comparison with a competing model, a higher-order closure), approved the written research plan, and reviewed the changes and the text. The code is verified by an automated test suite run in continuous integration: exact log-likelihoods against closed forms and independent computations (including a Hawkes process), exact stationary distributions of small models against long simulations, analytic gradients against finite differences, replayed against simulated log-likelihoods, time-rescaling residuals of simulated data against the unit exponential distribution, and, in debug mode, the recomputation of every rate after every event; the separate implementation of the competing model agrees with tricl to 10^{-12} where the models coincide. Every number, table and figure in the paper is produced by scripts from the code and the public data, so that it can be regenerated and checked. The author takes full responsibility for the content of the paper.

ACKNOWLEDGMENTS

[To be completed by the author.]

Appendix A: The log-likelihood

Let $x(t)$ be the state at time t: the set of links, and the impact $I_a(t)$ of every existing action a. Between events, impacts decay deterministically, $I_a(t) = I_a(t_0)\, e^{-\delta_a (t-t_0)}$, and an action ceases to exist when its impact reaches its threshold, at a deterministic time. Forced events happen at given times, and triggered acts happen together with the contact events that trigger them. All other events are random with rates $\lambda_e(x(t))$ that depend on the links and, through angles weighted by impacts, continuously on time. Let $\Lambda(t)$ be their total rate.

Proposition 1. *Given the initial state and the forced events, the probability density of a sequence of random events $e_1, \dots, e_m$ at times $t_1 < \dots < t_m$ in $[0, T]$ is*

$$p = \prod_{i=1}^{m} \lambda_{e_i}(x(t_i^-)) \, \exp\Big(-\int_0^T \Lambda(t)\, dt\Big),$$

and its logarithm is Eq. (2).

Proof. Between the times of random events, forced events, and expiries of actions, the state changes only deterministically, so the process is a point process with the stochastic intensity $\Lambda(t)$ with respect to its history, and an event at time t is e with probability $\lambda_e(x(t^-))/\Lambda(t)$. The density of a point process with intensity Λ is $\prod_i \Lambda(t_i) \exp(-\int_0^T \Lambda\, dt)$, and multiplying by the probabilities of the identities gives p. Forced events and expiries happen at times determined by the past and the given input, and triggered acts with probability one, so they add no factors. □

tricl computes the integral event by event: the rate of an event changes deterministically between its updates, and when an event is updated or removed, the integral of its rate since its last update is added by Gauss–Legendre quadrature (eight nodes on pieces of length at most $1/\delta_{\max}$, where $\delta_{\max}$ is the largest decay rate of its weighted angles, which gives a relative accuracy of about 10^{-13}). The gradient is accumulated in the same way. For the symmetric relationship type of contacts, the two directions $e = (i, j)$ and $\bar{e} = (j, i)$ of a pair are two possible events that cause the same change; Proposition 1 is the likelihood of the sequence with the recorded direction of each event, and the likelihood without directions replaces λ_{e_i} by $\lambda_{e_i} + \lambda_{\bar{e}_i}$. Since all models in this paper have $\lambda_e = \lambda_{\bar{e}}$ in every state, the two log-likelihoods differ by the constant $m \ln 2$, and estimates, likelihood differences and standard errors are the same. The competing model C1 is implemented with the same convention.

Appendix B: Exactness of the simulation

The sum tree of rates holds, for every individually scheduled event e, a weight $w_e \geq \lambda_e(x(s))$ for all times s until its next update: for constant rates $w_e = \lambda_e$, and for rates with decaying impacts $w_e = \bar{A} f(\bar{B})$, with $\bar{A}$ the attempt rate at the last update (attempt coefficients are non-negative, so the attempt rate decreases as impacts decay) and $\bar{B}$ the probability units at the last update with all negative contributions of decaying angles omitted. Pairs without adjacent angles are covered by one entry per event type, whose weight is an upper bound of the sum of their rates; an attempt of this entry draws a pair through a channel chosen proportionally to its attempt rate (a uniformly random pair, or a uniformly random link of a leg type and a uniformly random partner) and is accepted with the ratio of the pair's success probability to an upper bound. Let W be the total weight.

Proposition 2. *If an attempt waits an exponential time with rate W, chooses an entry proportionally to its weight, performs a chosen individually scheduled event e with probability $\lambda_e(x(t))/w_e$ (and otherwise refreshes its weight at time t), and processes forced events and expiries when their times are reached before the next attempt, then the produced events have the law of the process of Proposition 1.*

Proof. Between updates, the attempts form a Poisson process of rate W whose points are independently marked; an attempt at time t produces event e with probability $\lambda_e(x(t))/W$, both for individually scheduled events and for covered pairs, since the channel probabilities and acceptance ratios multiply to λ_e/W. By the thinning theorem [10, 11], the produced events form a point process with intensity $\lambda_e(x(t))$ for each e, as long as $w_e \geq \lambda_e(x(s))$ on the interval, which holds by construction. Refreshing a weight after a rejected attempt and processing deterministic events changes the bounds and the state only at stopping times, after which the argument applies anew with fresh exponential waiting times (the waiting time is memoryless). □

Appendix C: Time-rescaling residuals

Under a correct model, the termination of a contact between i and j that started at time s has the hazard $h(t) = \lambda_{(i,j)}(x(t)) + \lambda_{(j,i)}(x(t))$ given the path, so that $\Lambda = \int_s^{s+\tau} h(t)\, dt$, with τ the duration, is a unit exponential variable, and these variables are independent across contacts [8]. tricl accumulates the integrated rate of every termination event from the time it is added (when the contact starts) to its occurrence, including time-varying rates, and adds the integral of the mirrored event of the same pair. Contacts still in progress at the end of a day are not included, which slightly favors short durations; the effect is small, since few contacts are in progress at 17:00.

Appendix D: Pair closure

For a class with N students and $M = N(N-1)/2$ pairs, let L_c be the number of pairs with marks c in contact (out of M_c), $L = \sum_c L_c$, $U = M - L$, and Z and O the sums of the numbers n of common contacts over pairs in contact and not in contact. Since every two-path is counted once in $Z + O$, $\sum_i d_i(d_i - 1) = 2(Z + O)$ for the numbers of contacts d_i. The closure assumes that, for a pair in contact, $n \sim \text{Poisson}(Z/L)$ and the other contacts of the two students beyond the common ones are Poisson with mean $2b_1$, $b_1 = (Z + O)/L - Z/L$ (the mean number of other contacts of an endpoint of a contact, $\sum_i d_i(d_i - 1)/\sum_i d_i$, minus the common ones), and for a pair not in contact $n \sim \text{Poisson}(O/U)$ with $b_0 = [\sum_i d_i(N - 1 - d_i)]/(2U) - O/U$. With $\lambda_c^{\pm}(n, m)$ the pair rates of formation and termination (both directions) for a pair with marks c, n common contacts and m other contacts of its endpoints, and $\langle\cdot\rangle_0$, $\langle\cdot\rangle_1$ the expectations under these distributions,

$$\begin{aligned}
\dot{L}_c &= (M_c - L_c)\langle\lambda_c^+\rangle_0 - L_c\langle\lambda_c^-\rangle_1, \\
\dot{Z} &= \textstyle\sum_c \big[(M_c - L_c)\langle 3n\lambda_c^+\rangle_0 - L_c\langle 3n\lambda_c^-\rangle_1\big], \\
\dot{O} &= \textstyle\sum_c \big[(M_c - L_c)\langle (m-n)\lambda_c^+\rangle_0 - L_c\langle (m-n)\lambda_c^-\rangle_1\big],
\end{aligned}$$

since a new contact with n common contacts turns its own n open two-paths into closed ones and adds two closed two-paths for each common contact ($\Delta Z = 3n$), and adds one open two-path for each other contact of its endpoints ($\Delta O = m - n$), and the termination of a contact reverses these changes. The simple closure replaces the conditional distributions by the same Poisson distributions for both states, with means $(N-2)\rho^2$ and $(N-2)\rho$, $\rho = L/M$.

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